

Anesthetics

Before the mid-19th century, surgery was very painful because it had to be carried out without effective painkillers. In the 1840s, the use of anesthetics—medicines that cause temporary loss of sensation—began to become widespread. Today, a wide range of anesthetics has made surgery largely pain free.

ANCIENT SURGERY

There is evidence of surgical operations from prehistoric times. One of the most widespread was trepanning—drilling holes in the head with sharp tools to relieve pain, in ritual ceremonies, or, as some believe, to let out “evil spirits.” This trepanned skull from the Middle East is more than 4,000 years old.

*Trepanning
hole in skull*



SPEEDY SURGERY

This 17th-century painting shows a dentist pulling a tooth. Before anesthetics, medical practitioners tried to make these procedures less painful by giving patients alcohol to drink—or sometimes even physically knocking them out—before operating on them as quickly as possible.

ETHER MIRACLE

The American dentist William Morton made a medical leap in 1846 when he used the chemical ether as an anesthetic on a patient before performing a pain-free operation. Ether wasn't without its problems, as it was highly flammable, but its use soon became widespread.



*A regulator
controls the flow
of ether fumes
into the tube.*

*A rubber tube carries
ether fumes from the
jar to the mouthpiece.*

**An early ether inhaler, based
on Morton's original design**